 

# AHTARI MYLLYMAKI, Finland

WMO: **029240**

Lat: **62.535N** Lon: **24.217E** Elev: **160** StdP: **99.42** Time Zone: **2.00 (EUE)** Period: **94-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **2** | **-27.6** | **-24.3** | **-30.5** | **0.2** | **-27.6** | **-26.8** | **0.3** | **-24.2** | **4.3** | **-4.6** | **3.8** | **-5.0** | **0.2** | **40** | **0.589** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **10.6** | **26.0** | **17.5** | **24.1** | **16.6** | **22.3** | **15.8** | **19.2** | **23.5** | **18.0** | **21.9** | **16.9** | **20.8** | **1.6** | **200** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **17.6** | **12.9** | **20.8** | **16.4** | **11.9** | **19.7** | **15.3** | **11.1** | **18.4** | **55.2** | **23.5** | **51.3** | **21.9** | **47.9** | **20.7** | **24.0** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **4.1** | **3.5** | **3.2** | DB | **-30.3** | **28.3** | **4.1** | **2.1** | **-33.3** | **29.8** | **-35.8** | **31.0** | **-38.1** | **32.1** | **-41.1** | **33.6** | (4) |
| (5) | | | | WB | **-30.5** | **20.8** | **4.0** | **1.7** | **-33.4** | **22.0** | **-35.8** | **23.0** | **-38.1** | **24.0** | **-41.0** | **25.3** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | DBAvg | | **3.3** | **-7.9** | **-8.0** | **-4.2** | **2.2** | **7.9** | **12.7** | **15.6** | **14.0** | **8.9** | **3.4** | **-1.1** | **-4.7** | (6) |
| (7) | | DBStd | | **9.49** | **7.04** | **7.27** | **5.33** | **3.59** | **4.15** | **3.48** | **3.01** | **3.00** | **3.54** | **4.06** | **4.67** | **6.50** | (7) |
| (8) | | HDD10.0 | | **2897** | **556** | **505** | **441** | **234** | **91** | **12** | **1** | **4** | **61** | **207** | **332** | **454** | (8) |
| (9) | | HDD18.3 | | **5505** | **814** | **738** | **699** | **483** | **323** | **172** | **97** | **138** | **282** | **464** | **582** | **713** | (9) |
| (10) | | CDD10.0 | | **451** | **0** | **0** | **0** | **1** | **26** | **93** | **173** | **127** | **29** | **1** | **0** | **0** | (10) |
| (11) | | CDD18.3 | | **17** | **0** | **0** | **0** | **0** | **0** | **3** | **11** | **3** | **0** | **0** | **0** | **0** | (11) |
| (12) | | CDH23.3 | | **233** | **0** | **0** | **0** | **0** | **16** | **51** | **131** | **36** | **0** | **0** | **0** | **0** | (12) |
| (13) | | CDH26.7 | | **29** | **0** | **0** | **0** | **0** | **1** | **5** | **21** | **2** | **0** | **0** | **0** | **0** | (13) |
| (14) | Wind | WSAvg | | **1.3** | **1.2** | **1.2** | **1.4** | **1.3** | **1.4** | **1.3** | **1.2** | **1.1** | **1.2** | **1.4** | **1.4** | **1.4** | (14) |
| (15) | Precipitation | PrecAvg | | **613** | **42** | **32** | **31** | **32** | **44** | **66** | **84** | **74** | **57** | **58** | **48** | **44** | (15) |
| (16) | | PrecMax | | **813** | **89** | **67** | **48** | **67** | **96** | **116** | **147** | **162** | **136** | **121** | **95** | **81** | (16) |
| (17) | | PrecMin | | **469** | **9** | **1** | **3** | **4** | **7** | **20** | **21** | **23** | **15** | **12** | **3** | **14** | (17) |
| (18) | | PrecStd | | **84** | **20** | **17** | **12** | **18** | **23** | **25** | **33** | **29** | **34** | **30** | **22** | **16** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **3.5** | **4.7** | **9.4** | **17.8** | **25.4** | **27.5** | **29.0** | **26.2** | **20.3** | **13.3** | **9.0** | **6.1** | (19) |
| (20) | | | MCWB | | **2.5** | **2.5** | **4.6** | **9.9** | **15.0** | **17.6** | **19.8** | **18.2** | **15.3** | **10.7** | **8.6** | **5.6** | (20) |
| (21) | | 2% | DB | | **2.2** | **2.7** | **6.2** | **14.0** | **22.3** | **24.9** | **26.6** | **24.2** | **18.2** | **11.5** | **7.4** | **4.6** | (21) |
| (22) | | | MCWB | | **1.6** | **1.7** | **2.7** | **7.7** | **13.4** | **16.2** | **18.3** | **17.2** | **14.2** | **10.3** | **6.7** | **4.2** | (22) |
| (23) | | 5% | DB | | **1.3** | **1.4** | **4.2** | **11.4** | **19.5** | **22.5** | **24.6** | **22.3** | **16.4** | **10.4** | **6.0** | **3.3** | (23) |
| (24) | | | MCWB | | **0.9** | **0.9** | **1.6** | **6.2** | **12.0** | **14.8** | **17.3** | **16.5** | **13.2** | **9.4** | **5.5** | **2.8** | (24) |
| (25) | | 10% | DB | | **0.4** | **0.5** | **2.5** | **9.2** | **16.7** | **20.5** | **22.7** | **20.5** | **14.9** | **9.3** | **4.7** | **2.1** | (25) |
| (26) | | | MCWB | | **0.1** | **0.0** | **0.7** | **5.1** | **10.4** | **14.0** | **16.7** | **15.7** | **12.5** | **8.4** | **4.3** | **1.7** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **2.8** | **2.9** | **5.1** | **10.9** | **16.2** | **19.1** | **21.2** | **20.1** | **16.2** | **11.6** | **8.5** | **5.7** | (27) |
| (28) | | | MCDB | | **3.1** | **3.9** | **8.6** | **14.9** | **22.1** | **24.9** | **26.9** | **24.1** | **19.0** | **12.4** | **8.9** | **5.9** | (28) |
| (29) | | 2% | WB | | **1.7** | **1.9** | **3.3** | **8.6** | **14.5** | **17.3** | **19.7** | **18.4** | **14.9** | **10.6** | **6.7** | **4.3** | (29) |
| (30) | | | MCDB | | **2.1** | **2.7** | **5.3** | **13.1** | **20.0** | **22.4** | **24.2** | **22.1** | **17.1** | **11.3** | **7.1** | **4.5** | (30) |
| (31) | | 5% | WB | | **1.0** | **1.0** | **2.1** | **6.9** | **12.9** | **16.0** | **18.6** | **17.4** | **14.0** | **9.6** | **5.6** | **2.8** | (31) |
| (32) | | | MCDB | | **1.2** | **1.4** | **3.7** | **10.8** | **18.4** | **20.5** | **22.7** | **20.9** | **15.8** | **10.3** | **6.0** | **3.2** | (32) |
| (33) | | 10% | WB | | **0.2** | **0.1** | **1.0** | **5.4** | **11.1** | **14.8** | **17.5** | **16.3** | **12.9** | **8.5** | **4.3** | **1.7** | (33) |
| (34) | | | MCDB | | **0.4** | **0.5** | **2.2** | **8.7** | **15.4** | **19.3** | **21.4** | **19.5** | **14.5** | **9.2** | **4.7** | **2.1** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **6.3** | **7.5** | **9.7** | **10.2** | **11.8** | **10.7** | **10.6** | **10.5** | **8.8** | **5.5** | **3.9** | **5.1** | (35) |
| (36) | | 5% DB | MCDBR | | **4.5** | **4.9** | **10.7** | **15.0** | **17.4** | **15.1** | **14.3** | **14.0** | **11.0** | **6.4** | **4.0** | **3.9** | (36) |
| (37) | | | MCWBR | | **4.4** | **4.3** | **7.7** | **9.1** | **9.2** | **7.5** | **6.8** | **7.5** | **7.2** | **5.1** | **4.0** | **3.8** | (37) |
| (38) | | 5% WB | MCDBR | | **4.5** | **4.7** | **8.8** | **13.7** | **15.4** | **12.1** | **12.0** | **11.6** | **9.4** | **5.7** | **4.1** | **3.8** | (38) |
| (39) | | | MCWBR | | **4.4** | **4.3** | **6.5** | **8.8** | **8.6** | **6.7** | **6.5** | **6.8** | **6.5** | **4.9** | **4.0** | **3.7** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.233** | **0.286** | **0.312** | **0.348** | **0.344** | **0.344** | **0.367** | **0.352** | **0.328** | **0.313** | **0.254** | **0.355** | (40) |
| (41) | | taud | | | **2.042** | **2.148** | **2.215** | **2.360** | **2.459** | **2.482** | **2.435** | **2.467** | **2.527** | **2.404** | **2.141** | **2.624** | (41) |
| (42) | | Ebn at Noon | | | **419** | **653** | **786** | **838** | **875** | **881** | **846** | **829** | **784** | **637** | **393** | **256** | (42) |
| (43) | | Edh at Noon | | | **34** | **66** | **91** | **98** | **96** | **96** | **99** | **88** | **69** | **53** | **30** | **20** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **0.19** | **0.74** | **2.23** | **3.64** | **4.87** | **5.24** | **4.97** | **3.83** | **2.22** | **0.89** | **0.23** | **0.09** | (44) |
| (45) | | RadStd | | | **0.03** | **0.17** | **0.25** | **0.44** | **0.54** | **0.43** | **0.51** | **0.42** | **0.27** | **0.18** | **0.04** | **0.01** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (49 neighbors) | **+0.49** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **-135** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page